

NoHeadcountSystem

High-Performance Cryptocurrency HFT System

Your Name / Team Name

Your Institution / Department

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香港中文大學 (深圳)
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- ① Team Division
- ② Project Background & Introduction
- ③ Core Architecture
- ④ Infrastructure
- ⑤ Strategies & Execution Layer
- ⑥ Risk Management
- ⑦ Backtest Engine
- ⑧ Dashboard & Deployment

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Team Division

表 1: Project Task Allocation

Team Member	Contribution
Mo Yunjie	Strategies, System Outline, Report
Li Taiyan	Infrastructure, OMS, GUI
Guo Xiaoyou	Event-Driven Backtesting
Huang Yujin	Risk Management, TCA

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Project Background & Introduction

- **System Positioning:** A High-Frequency Trading (HFT) system tailored for futures and cryptocurrencies.
- **Core Features:**
 - **Modular Design:** Decoupled infrastructure, strategies, risk management, and backtesting components.
 - **Event-Driven:** Non-blocking data and instruction routing based on a unified event bus.
 - **Seamless Switching:** One-click switch between simulated matching (SIM) and Binance Spot Testnet (BINANCE).
- **Tech Stack:** Python, Numba (High-performance computing), uv (Fast dependency management).

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Core Architecture & Data Flow

- **System Bus:** Modules are fully decoupled using the EventEngine.
- **Data Pipeline:**
 - ① **Tick:** Market data enters the system.
 - ② **Strategy:** Listens to Ticks and generates a Signal.
 - ③ **OMS (Order Management):** Converts Signals into standardized Orders.
 - ④ **RMS (Risk Management):** Intercepts and validates orders (limits, exposure).
 - ⑤ **Gateway:** Routes to the target exchange or simulator.
 - ⑥ **Trade:** Execution reports flow back via the event bus to update all modules.

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Infrastructure Layer

- **Gateway Connection:**
 - `BinanceTestnetGateway`: Connects to Binance Spot Testnet for live testing.
 - `SimulatedGateway`: Local simulated matching engine for safe debugging.
- **Unified Data Models**: Standardized `TickData`, `Signal`, `OrderData`, `TradeData`.
- **Order Management System (OMS)**:
 - Bridges the strategy end (signals) and the channel end (orders).
 - Automatically tracks order status updates and maintains the signal-to-order mapping lifecycle.

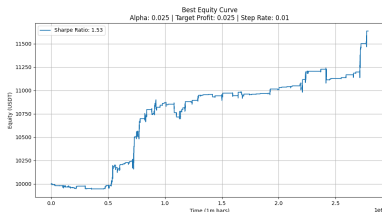
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Strategies & Execution Layer

- **Strategy Base (base.py):** Standardized interfaces where strategies only emit signals without placing orders directly, keeping responsibilities single and clean.
- **Featured Strategy: FeeAwareStatArbStrategy (dca.py)**
 - **Statistical Arbitrage:** Uses an Exponential Moving Average (EMA) to calculate micro-price deviations.
 - **Fee-Aware Logic:** Entries require the potential profit to strictly exceed double taker fees ($\text{taker_fee} \times 2$) plus a target profit rate.
 - **Dollar-Cost Averaging (DCA):** Dynamically adds to losing positions with strict step rates to lower the average cost.
 - **Smart Order Splitting:** Automatically splits position adjustments into legal CLOSE and OPEN offset signals.

Arbitrage Strategy Results

In-Sample Equity Curve



Out-of-Sample Metrics

- Initial Capital: 10000.0
- Realized PnL: 66.20
- Turnover: 9770.01
- End Position: 0.0
- Closed Trades: 8
- **Annualized Return:**
0.66%
- **Annualized Volatility:**
0.47%
- **Sharpe Ratio:** 1.42
- **Max Drawdown:** 0.37%

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Risk Management Center (RMS)

- **Real-time Interception System:** Acts as the last line of defense before orders are dispatched to the exchange (`manager.py`).
- **Core Risk Rules:**
 - **Max Order Volume** (`max_order_volume`): Prevents fat-finger errors and extreme anomalies.
 - **Max Net Position** (`max_symbol_position`): Controls unilateral exposure and inventory risks.
- **Dynamic Scalability:** The architecture allows for hot-plugging of new risk validation rules without altering the core pipeline.

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High-Performance Backtest Engine

- **Core Driver:** Numba-accelerated K-line backtesting core (exchange.py & loop.py).
- **Matching Mechanism:**
 - **Bar-Level Dual Sweep:** Simulates intra-bar volatility by scanning triggering limits through High and Low prices sequentially.
 - **Aggressive Limit Conversion:** Automatically converts marketable limit orders to market orders.
- **Accurate Accounting:**
 - Built-in Taker/Maker commission models.
 - Real-time maintenance of positions, cost prices, realized/unrealized PnL, and capital turnover.

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Dashboard & GUI

- **GUI Monitor:** Real-time metrics, dynamic grids for orders/trades/ticks, and event timeline tracking.
- Supports headless mode (*ENABLE_GUI=0*) for cloud server deployment.

Algo Trading Order Monitor									
订单: 8	成交: 5	成交完成: 5	拒单: 0	Tick: 15					
订单监控									
订单ID	策略	合约	状态	价格	数量	已成交	拒单原因	时间	
LOCAL_acd0162943	pulse_buy_v1	BTCSUOT.BINANCE	SUBMITTED	68176.66	0.0001	0.0		14:16:25	
LOCAL_94241083db	pulse_buy_v1	BTCSUOT.BINANCE	SUBMITTED	68171.9	0.0001	0.0		14:16:21	
LOCAL_9b41f2467a	pulse_buy_v1	BTCSUOT.BINANCE	SUBMITTED	68171.9	0.0001	0.0		14:16:18	
LOCAL_095204cf21	pulse_buy_v1	BTCSUOT.BINANCE	FILLED	68171.9	0.0001	0.0001		14:16:13	
LOCAL_2ef73ff1d6	pulse_buy_v1	BTCSUOT.BINANCE	FILLED	68171.9	0.0001	0.0001		14:16:10	
LOCAL_d8fd7efafa	pulse_buy_v1	BTCSUOT.BINANCE	FILLED	68188.76	0.0001	0.0001		14:16:07	
LOCAL_2f289b17d1	pulse_buy_v1	BTCSUOT.BINANCE	FILLED	68188.76	0.0001	0.0002		14:16:00	
LOCAL_c7667f1f06	pulse_buy_v1	BTCSUOT.BINANCE	FILLED	68188.76	0.0001	0.0001		14:15:55	
成交监控									
成交ID	订单ID	策略	合约	价格	数量	方向	时间		
BINANCE_258942	LOCAL_095204cf21	pulse_buy_v1	BTCSUOT.BINANCE	66835.2	0.0001	LONG	14:16:28		
BINANCE_258940	LOCAL_2ef73ff1d6	pulse_buy_v1	BTCSUOT.BINANCE	66835.2	0.0001	LONG	14:16:23		
BINANCE_258936	LOCAL_d8fd7efafa	pulse_buy_v1	BTCSUOT.BINANCE	66835.2	0.0001	LONG	14:16:19		
BINANCE_258924	LOCAL_2f289b17d1	pulse_buy_v1	BTCSUOT.BINANCE	66851.73	0.0001	LONG	14:16:11		
BINANCE_258919	LOCAL_c7667f1f06	pulse_buy_v1	BTCSUOT.BINANCE	66851.73	0.0001	LONG	14:16:07		
Tick监控									
合约	最新价	买一	卖一	时间					
BTCSUOT.BINANCE	66839.855000000001	66839.85	66839.86	14:16:29					
BTCSUOT.BINANCE	66839.855000000001	66839.85	66839.86	14:16:27					
BTCSUOT.BINANCE	66839.855000000001	66839.85	66839.86	14:16:25					
BTCSUOT.BINANCE	66835.195	66835.19	66835.2	14:16:21					
BTCSUOT.BINANCE	66835.195	66835.19	66835.2	14:16:20					
BTCSUOT.BINANCE	66835.195	66835.19	66835.2	14:16:18					
BTCSUOT.BINANCE	66835.195	66835.19	66835.2	14:16:15					
BTCSUOT.BINANCE	66835.195	66835.19	66835.2	14:16:13					

Quick Start

- **Minimal Dependencies:** Uses `uv` (Rust-based Python package manager) for instant environment syncing (`uv sync`).
- **Config-Driven (`.env`):**
 - `TRADING_MODE=SIM/BINANCE` for one-click environment switches.
 - Centralized configuration for API keys, proxies, and strategy parameters.
- **One-Command Run:** `uv run src/main.py` automatically initializes all modules and starts the event loop.

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Conclusion & Future Work

- **System Advantages:**

- Clear isolation of responsibilities (signals decoupled from orders).
- High extensibility for new exchanges and strategies.
- Consistent logic between backtesting and live trading.

- **Future Extensions:**

- Integrate more exchange gateways (e.g., OKX, Bybit).
- Introduce Machine Learning models for alpha signal prediction.
- Enhanced backtest charting and detailed attribution reports.

Q&A / Thank You!